MA 102 (Mathematics II - Ordinary Differential Equations)

Department of Mathematics, IIT Guwahati
Semester: January-April, 2024

Solution of Tutorial Sheet No. 4 Date of Discussion: April 12, 2024

Lectures 10-14: Singular points, Power series solution for second-order ODE, Bessel functions, Legendre polynomials

---

Attention students: Please keep in mind that it is not possible to discuss all problems during the tutorial. You can consider the problems not discussed as your practice problems.

1. Determine the radii of convergence of the following power series:

   (a) $\displaystyle\sum_{n=0}^{\infty} \frac{n^2}{2^n}(x+2)^n$; (b) $\displaystyle\sum_{n=1}^{\infty} \frac{3}{n^3}(x-2)^n$; (c) $\displaystyle\sum_{n=0}^{\infty} \frac{2^{-n}}{n+1}(x-1)^n$.
   Solution:
   The radii of convergence can be easily found by using ratio test: (a) $R = 2$, (b) $R = 1$, (c) $R = 2$.

2. Classify the singular points of the following differential equations:

   (a) $e^x y'' - (x^2-1)y' + 2xy = 0$; (b) $(x^2+x)y'' + 3y' - 6xy = 0$;
   (c) $(\sin x)y'' + (\cos x)y = 0$; (d) $\ln(x-1)y'' + (\sin 2x)y' - e^x y = 0, \quad$ for $x > 1$.
   Solution:
   (a) The given differential equation can be written as

   $$y'' - (x^2-1)e^{-x}y' + 2xe^{-x}y = 0.$$

   Here, $P(x) = -(x^2-1)e^{-x}, \;\; Q(x) = 2xe^{-x}$. Both are analytic at all $x \in \mathbb{R}$. Therefore, all $x \in \mathbb{R}$ are ordinary points. Hence, this differential equation does not have any singular point.
   (b) The given differential equation can be written as

   $$y'' + \frac{3}{x(x+1)}y' - \frac{6}{x+1}y = 0.$$

   Here, $P(x) = \dfrac{3}{x(x+1)}, \;\; Q(x) = -\dfrac{6}{x+1}$. It is obvious that $x = 0$ and $x = -1$ are the singular points. Using the expressions $(x-x_0)P(x)$ and $(x-x_0)^2Q(x)$ for both $x = 0$ and $x = -1$, it can be found that both are regular singular points.
   (c) The given differential equation can be written as

   $$y'' + \frac{\cos x}{\sin x}y = 0.$$

Here, $P(x)=0, \quad Q(x)=\dfrac{\cos x}{\sin x}$. Hence, $Q(x)$ is not analytic at $x=n\pi,\ n\in\mathbb{Z}$. Therefore, $\{n\pi, n\in\mathbb{Z}\}$ is the set of singular points.
Since $\lim\limits_{x\to n\pi}(x-n\pi)\dfrac{\cos x}{\sin x}$ exists, $x=n\pi$ are regular singular points for all n.
(d) The given differential equation can be written as

$$y''+\frac{\sin 2x}{\ln(x-1)}y'-\frac{e^x}{\ln(x-1)}y=0.$$

Here, $P(x)=\dfrac{\sin 2x}{\ln(x-1)}, \quad Q(x)=-\dfrac{e^x}{\ln(x-1)}$. Hence, $Q(x)$ is not analytic at $x=2$, making it a singular point. Check it to get it as a regular singular point.

3. Find the indicial equation and their roots of the following differential equations:

(a) $(x^2-x-2)y''+(x^2-4)y'-6xy=0$; (b) $x^2y''+2(x-3x^2)y'+e^xy=0$;
(c) $x^2y''+(\sin x)y'+(\cos x)y=0$.
Solution:
(a) The given differential equation can be written as

$$y''+\frac{x+2}{x+1}y'-\frac{6x}{(x-2)(x+1)}y=0.$$

It is obvious that $x=-1$ and $x=2$ are regular singular points. We know that the indicial equation is given by

$$m(m-1)+mp_0+q_0=0.$$

Now, since we have two regular singular points, the indicial equation will be different corresponding to each singular point.
The indicial equation in the neighbourhood of $x=2$ is

$$m(m-1)=0 \quad \Rightarrow m=0,\ m=1.$$

This is because

$$p_0=\lim_{x\to 2}(x-2)\frac{(x+2)}{x+1}=0,$$
$$q_0=\lim_{x\to 2}(x-2)^2\frac{(x+2)(-6x)}{(x+1)(x+2)}=0.$$

Similarly, in the neighbourhood of $x=-1$, we have

$$p_0=\lim_{x\to -1}(x+1)\frac{(x+2)}{x+1}=1,$$
$$q_0=\lim_{x\to -1}(x+1)^2\frac{(-6x)}{(x+1)(x+2)}=0.$$

Hence, the indicial equation in the neighborhood of $x=-1$ is

$$m(m-1)+m+0=0 \quad \Rightarrow m=0,\ m=0.$$

(b) The given differential equation can be written as

$$y'' + 2\left(\frac{1-3x}{x}\right)y' + \frac{1+x+x^2/2!+\cdots}{x^2}y = 0.$$

It is clear that $x=0$ is a regular singular point and also that $p_0 = 2,\ q_0 = 1$. The indicial equation is $m(m-1)+2m+1 = 0$ or, $m^2+m+1 = 0$ which gives $m = -\frac{1}{2} \pm \frac{3\mathrm{i}}{2}$.
(c) The given differential equation can be written as

$$y'' + \frac{\sin x}{x^2}y' + \frac{\cos x}{x^2}y = 0.$$

By expressing $\sin x$ and $\cos x$ as infinite series x, we can rewrite the differential equation as

$$y'' + \frac{x - x^3/3! + x^5/5! + \cdots}{x^2}y' + \frac{1 - x^2/2! + x^4/4! + \cdots}{x^2}y = 0.$$

It is clear that $x=0$ is a regular singular point and also that $p_0 = 1,\ q_0 = 1$. The indicial equation is $m(m-1)+m+1 = 0$ or, $m^2+1 = 0$ which gives $m = \pm\mathrm{i}$.

4. Find a series solution about the ordinary point of each of the following differential equations subject to the given initial conditions:

(a) $(4-x^2)y'' + 2y = 0,\ \ y(0) = 0,\ y'(0) = 1;$
(b) $y'' - x^2y' + y\sin x = 0,\ \ y(0) = 0,\ y'(0) = 1.$
Solution:
(a) The given equation is $(4-x^2)y''+2y = 0$. It is clear that $x=0$ is an ordinary point. Hence, we can assume a solution of the form

$$y = \sum_{k=0}^{\infty} a_k x^k.$$

Finding the expression of y'' and putting it, along with y, back into the equation, we get

$$y = \sum_{k=2}^{\infty} a_k k(k-1)x^{k-2} - \sum_{k=2}^{\infty} a_k k(k-1)x^k + 2\sum_{k=0}^{\infty} a_k x^k = 0.$$

By making all series summations from 0 to ∞ and equating the coefficient of x^k, we get the recursion relation as

$$a_{k+2} = -\frac{2-k(k-1)}{4(k+2)(k+1)}a_k.$$

This gives us

$$\begin{aligned}
a_2 &= -\frac{2}{4\cdot 2}a_0 = \frac{1}{4}a_0\\
a_3 &= -\frac{2}{4\cdot 3\cdot 2}a_1 = -\frac{1}{12}a_1\\
a_4 &= 0\\
a_5 &= \frac{4}{4\cdot 5\cdot 4}a_3 = -\frac{1}{240}a_1\\
a_6 &= a_8 = \cdots = 0.
\end{aligned}$$

Using the initial condition $y(0) = 0$ gives $a_0 = 0$ and hence $a_2 = 0$.
Using the initial condition $y'(0) = 1$ gives $a_1 = 1$.
Hence the solution can be written as

$$\begin{aligned} y &= x - \frac{1}{12}x^3 - \frac{1}{240}x^5 - \cdots \\ &= x - 2\left[\frac{1}{3}\left(\frac{x}{2}\right)^3 + \frac{1}{15}\left(\frac{x}{2}\right)^5 + \cdots\right] \\ &= x - 2\sum_{n=1}^{\infty} \frac{1}{4n^2-1}\left(\frac{x}{2}\right)^{2n+1}. \end{aligned}$$

(It is not always necessary to write the solution in a series form.)
(b) The given equation is $y'' - x^2y' + y\sin x = 0$. It is clear that $x = 0$ is an ordinary point. Hence, we can assume a solution of the form

$$y = \sum_{k=0}^{\infty} a_k x^k.$$

Finding the expression of y'', y' and putting them, along with y, back into the equation, we get

$$\sum_{k=0}^{\infty}(k+2)(k+1)a_{k+2}x^k - \sum_{k=0}^{\infty}(k+1)a_{k+1}x^{k+2} + \sum_{k=0}^{\infty} a_k x^{k+1} - \sum_{k=0}^{\infty} a_k \frac{x^{k+3}}{3!} + \sum_{k=0}^{\infty} a_k \frac{x^{k+5}}{5!} + \cdots = 0,$$

where we have used the series expression of $\sin x = x - \dfrac{x^3}{3!} + \dfrac{x^5}{5!} + \cdots$. Using the given conditions, we get $a_0 = 0$ and $a_1 = 1$. Equating the coefficients of x^k, we get the following (infinite) multi-term recursion relation:

$$(k+2)(k+1)a_{k+2} + (2-k)a_{k-1} - \frac{a_{k-3}}{3!} + \frac{a_{k-5}}{5!} - \cdots = 0.$$

Now,

$$\begin{aligned} &k = 0 \Rightarrow a_2 = 0 \\ &k = 1 \Rightarrow a_3 = 0 \\ &k = 2 \Rightarrow a_4 = 0 \\ &k = 3 \Rightarrow a_5 = 0 \\ &k = 4 \Rightarrow (6 \cdot 5)a_6 - \frac{a_1}{3!} \Rightarrow a_6 = \frac{1}{180} \\ &k = 5 \Rightarrow a_7 = 0 \\ &k = 6 \Rightarrow (8 \cdot 7)a_8 + \frac{a_1}{5!} \Rightarrow a_8 = -\frac{1}{6720} \end{aligned}$$

and so on. The solution can be written as

$$y = x + \frac{x^6}{180} - \frac{x^8}{6720} + \cdots.$$

5. Find a series solution about the regular singular point of each of the the following differential equations:

(a) $xy'' + 4y' - xy = 0$; (b) $4x^2y'' + 2x^2y' - (x+3)y = 0$.

Solution: (a) For this differential equation, $x = 0$ is a regular singular point. Hence, we need to use Frobenius series solution:

$$y = \sum_{k=0}^{\infty} a_k x^{m+k}.$$

Putting y, y' and y'' in the differential equation, we get

$$\sum_{k=0}^{\infty} a_k(m+k)(m+k-1)x^{m+k-1} + 4\sum_{k=0}^{\infty} a_k(m+k)x^{m+k-1} - \sum_{k=0}^{\infty} a_k x^{m+k+1} = 0.$$

The indicial equation can be obtained as

$$m^2 + 3m = 0,$$

giving $m = 0, -3$.. The recursion relation can be obtained as

$$a_k = \frac{a_{k-2}}{(m+k)(m+k+3)}, \; k \geq 2.$$

Case 1: $m = 0$

The recursion relation is

$$a_k = \frac{a_{k-2}}{k(k+3)}, \; k \geq 2.$$

It can be found that $a_1 = 0$. The general expression for a_k is

$$a_{2k} = \frac{a_0}{2^n n!(2n+3)(2(n-1)+3)\cdots 7\cdot 5}.$$

Subsequently, the series solution can be obtained as

$$y = a_0\left(1 + \frac{x^2}{10} + \frac{x^4}{280} + \cdots\right).$$

Case 2: $m = -3$

The recursion relation is

$$a_k = \frac{a_{k-2}}{(k-3)k}, \; k \geq 2.$$

Proceed similarly as in Case 1.

(b) Proceed similarly as above by using Frobenius series solution. The indicial equation can be obtained as $4m^2 - 4m - 3 = 0$ giving $m = \frac{3}{2}, -\frac{1}{2}$. Then, proceed in the usual way to find the solution.

6. Use the orthogonality property of Legendre polynomials to show that $\int_{-1}^{1} g(x)P_n(x)dx = 0$ for every polynomial $g(x)$ with $\deg(g(x)) < n$.

Solution:

Since $g(x)$ is a polynomial of degree m with $m < n$, we can write $g(x) = \sum_{k=0}^{m} a_k P_k(x)$.

Therefore,

$$\int_{-1}^{1} g(x)P_n(x)\,dx = \sum_{k=0}^{m} a_k \int_{-1}^{1} P_k(x)P_n(x)\,dx = 0,$$

since $k \leq m < n$.

7. Find the specific solution to the differential equation $(1-x^2)y''-2xy'+12y=0$ subject to $y'(0)=4$ and the function $y(x)$ is well-behaved at $x=1$.
Solution: The given differential equation has the solution

$$y = AP_3(x) + BQ_3(x).$$

Since $Q_3(x)$ is not bounded when $x \to \pm 1$, we take $B=0$ for a bounded solution. Therefore, the solution can be written as

$$y = AP_3(x) = A\frac{1}{2}(5x^3 - 3x).$$

Using the given condition $y'(0)=4$, we get $A=-8/3$. Hence, the specific solution is

$$y = -\frac{4}{3}(5x^3 - 3x).$$

8. Using the substitution $\cos\theta = x$, show that the differential equation

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{dy}{d\theta}\right) + n(n+1)y = 0$$

can be converted to the standard form of Legendre equation.
Solution: The given differential equation can be rewritten as

$$\frac{d^2y}{d\theta^2} + \cot\theta\frac{dy}{d\theta} + n(n+1)y = 0.$$

Since $x=\cos\theta$, we have $\dfrac{dx}{d\theta} = -\sin\theta = -\sqrt{1-x^2}$. This gives us

$$\begin{aligned}\frac{dy}{d\theta} &= -\sqrt{1-x^2}\frac{dy}{dx},\\ \frac{d^2y}{d\theta^2} &= (1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx}.\end{aligned}$$

Putting these back in the given differential equation, we get

$$(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + n(n+1)y = 0.$$

9. Find a general solution to each of the following differential equations in terms of Bessel functions:

(a) $4x^2y'' + 4xy' + (4x^2-1)y = 0$; (b) $x^2y'' + xy' + x^2y = 0$;
(c) $x^2y'' + xy' + (16x^2 - 1/9)y = 0$; (d) $x^2y'' + xy' + ((4/9)x^2 - 9/4)y = 0$.
Solution:
(a) $y = c_1J_{1/2}(x) + c_2J_{-1/2}(x)$
(b) $y = c_1J_0(x) + c_2Y_0(x)$
(c) $y = c_1J_{1/3}(4x) + c_2J_{-1/3}(4x)$
(d) $y = c_1J_{3/2}((2/3)x) + c_2J_{-3/2}((2/3)x)$

10. The differential equation $x^2y'' + (x^2 - 2)y = 0, \quad x > 0$ does not look like a Bessel's equation. However, show that, by making a substitution $y = v(x)\sqrt{x}$ of the dependent variable, its solution can be expressed in terms of Bessel functions.

Solution: The given differential equation is $x^2y'' + (x^2 - 2)y = 0, \quad x > 0$. The given substitution is $y = v(x)\sqrt{x}$. This will lead to

$$\frac{dy}{dx} = x^{1/2}\frac{dv}{dx} + \frac{1}{2}x^{-1/2}v,$$
$$\frac{d^2y}{dx^2} = x^{1/2}\frac{d^2v}{dx^2} + x^{-1/2}\frac{dv}{dx} - \frac{1}{4}x^{-3/2}v.$$

Putting all these back in the differential equation and after some simplification, we get

$$x^2\frac{d^2v}{dx^2} + x\frac{dv}{dx} + \left(x^2 - \frac{9}{4}\right)v = 0,$$

which is Bessel's equation in the dependent variable z. Its solution is

$$v(x) = c_1J_{3/2}(x) + c_2J_{-3/2}(x).$$

Writing in the original variables,

$$y(x) = x^{1/2}(c_1J_{3/2}(x) + c_2J_{-3/2}(x)).$$

The End